

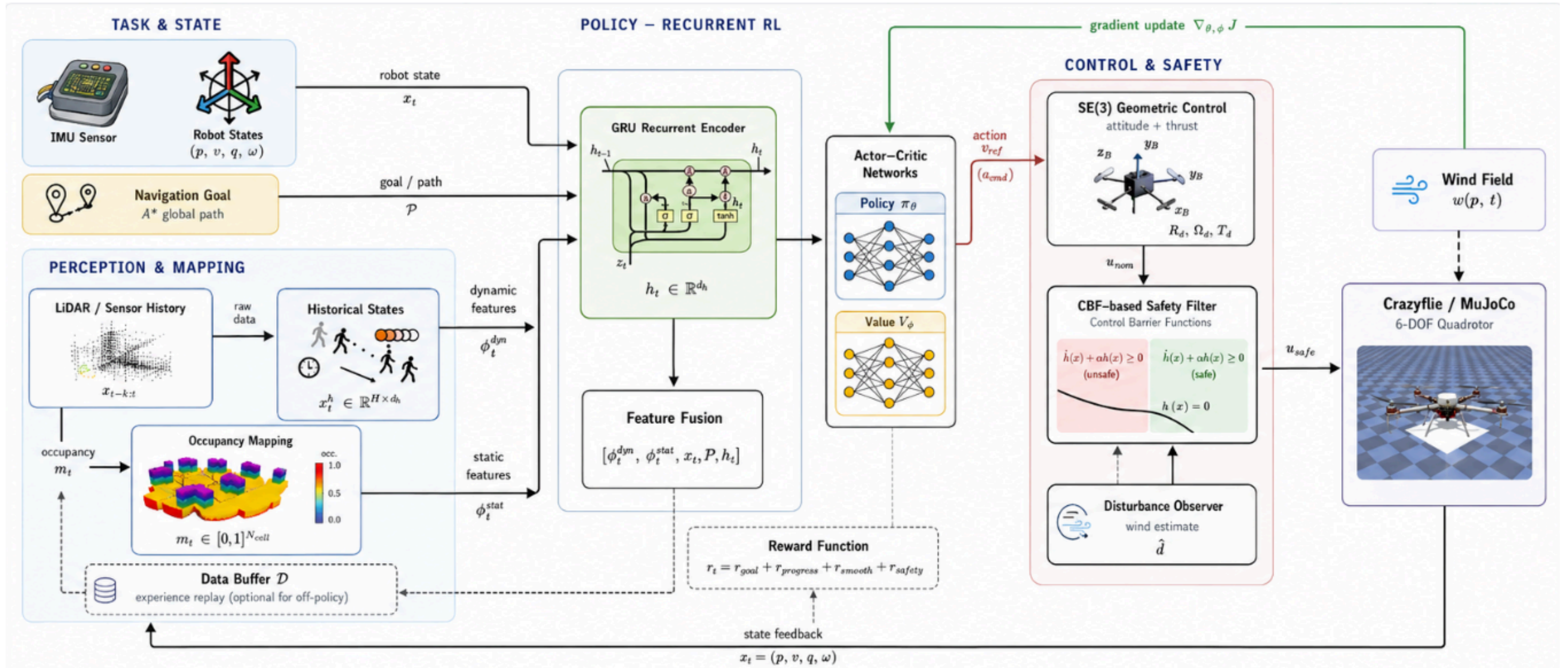
Safe Drone Navigation Under Disturbance

Presented by: Salem Fradi

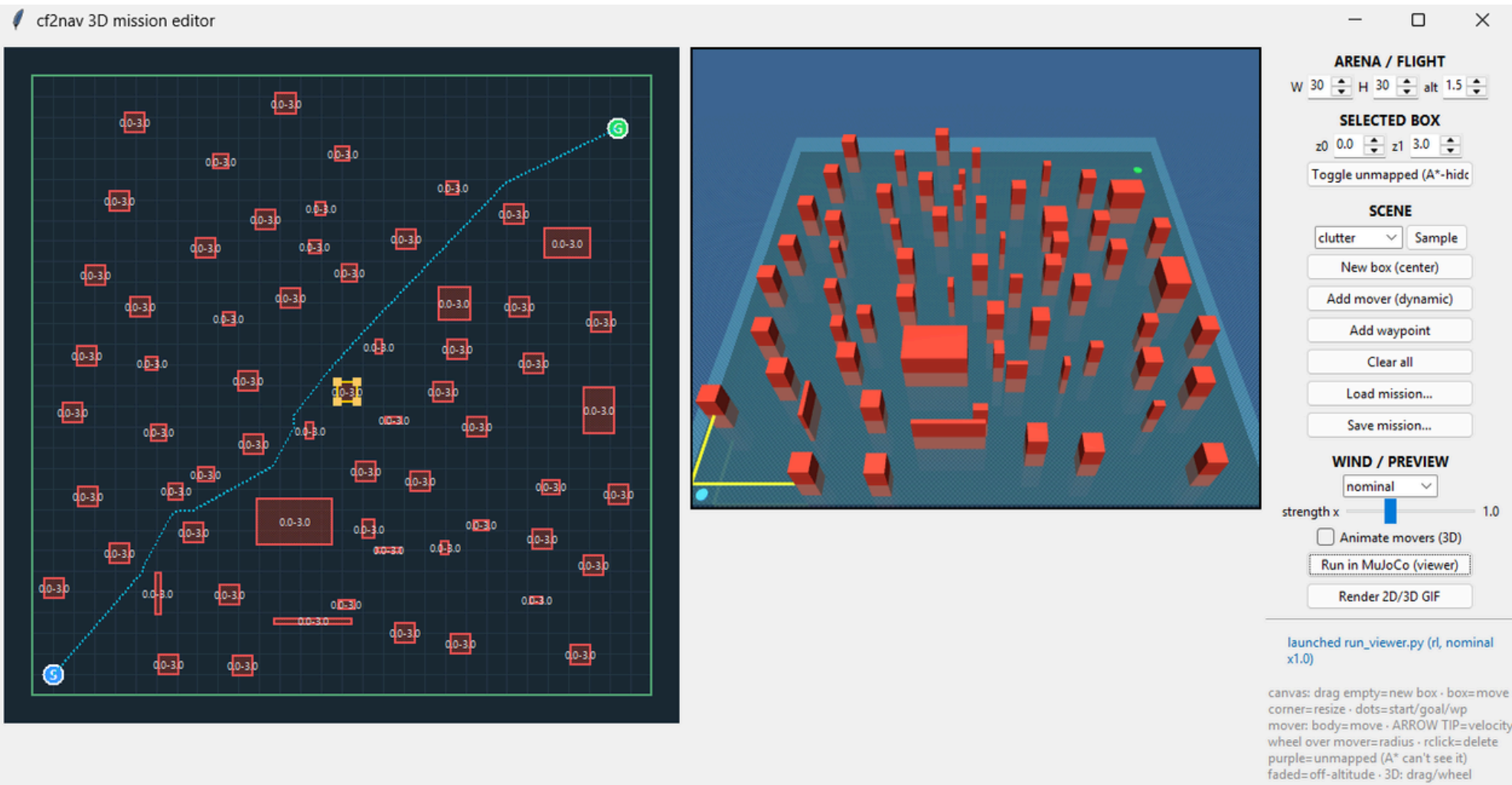
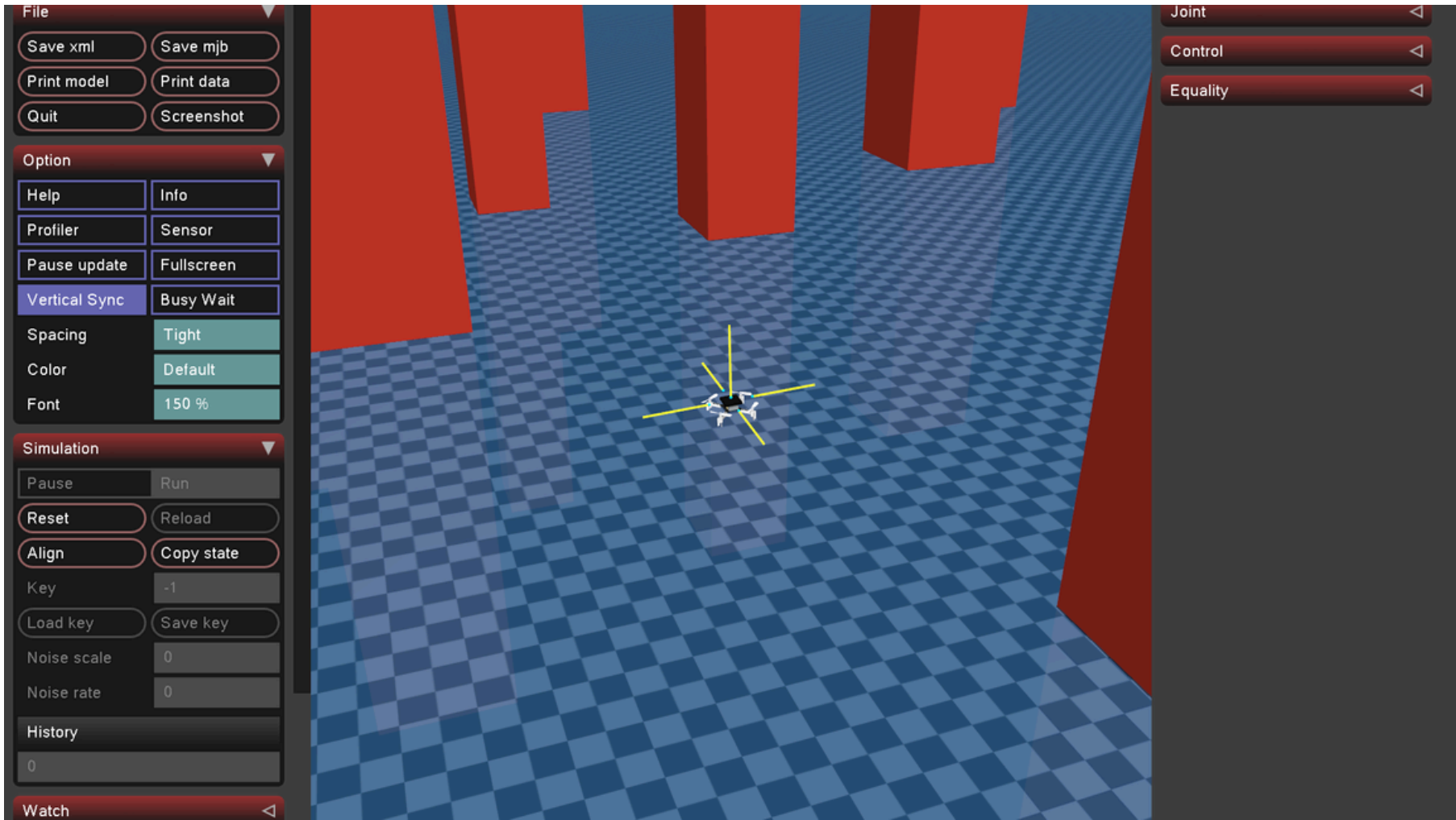
جامعة الملك عبد الله
للعلوم والتقنية
King Abdullah University of
Science and Technology



System Architecture:



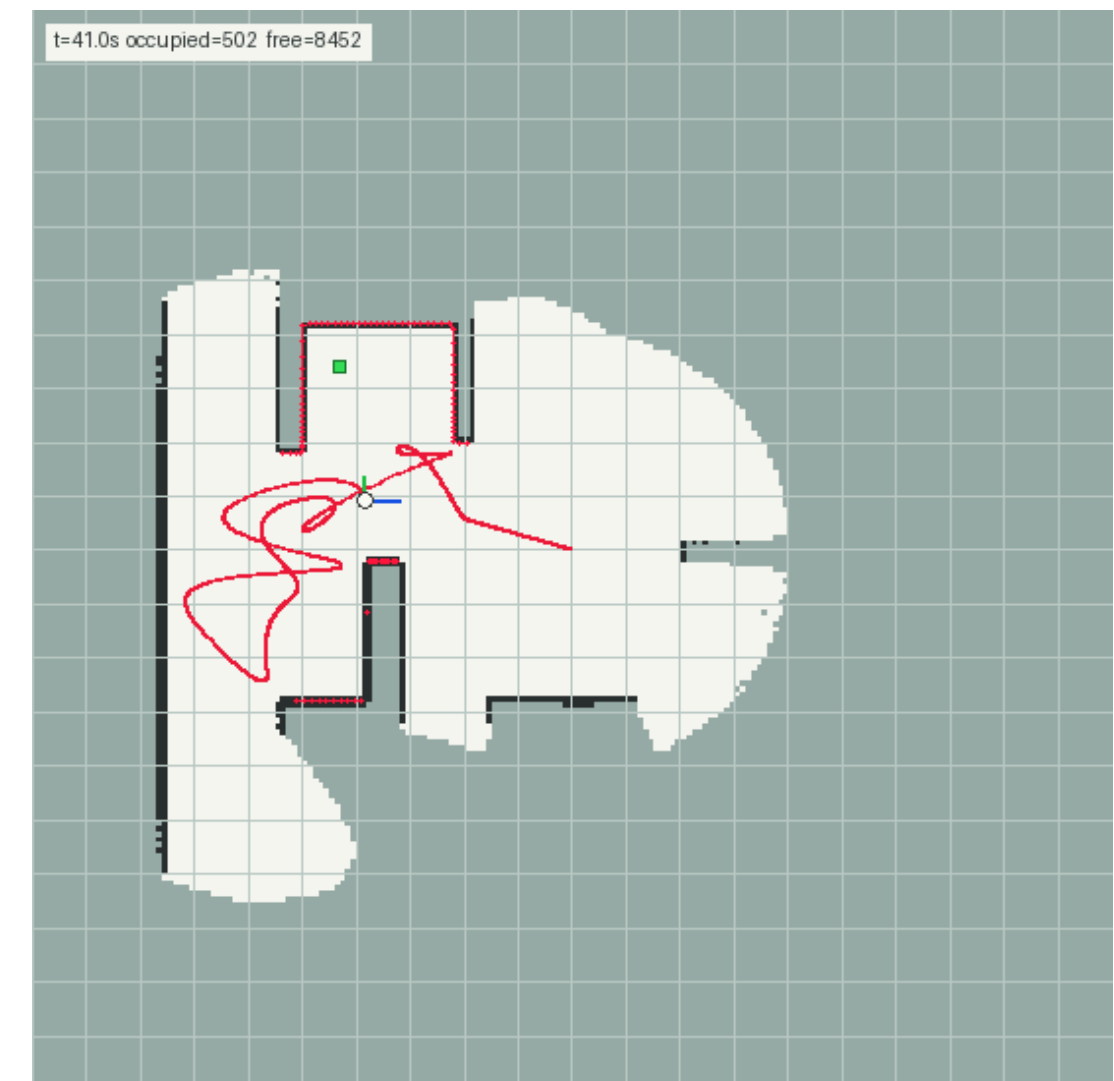
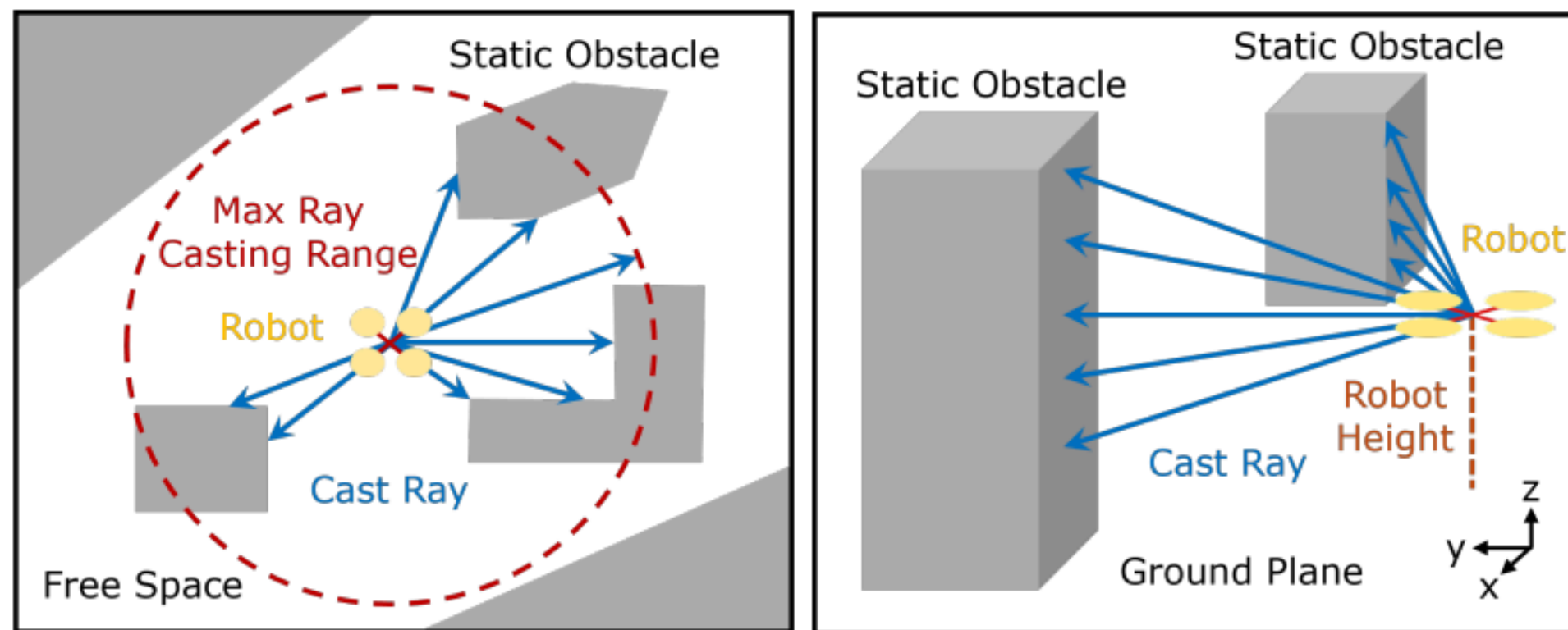
MuJoCo: Advanced Physics Simulation



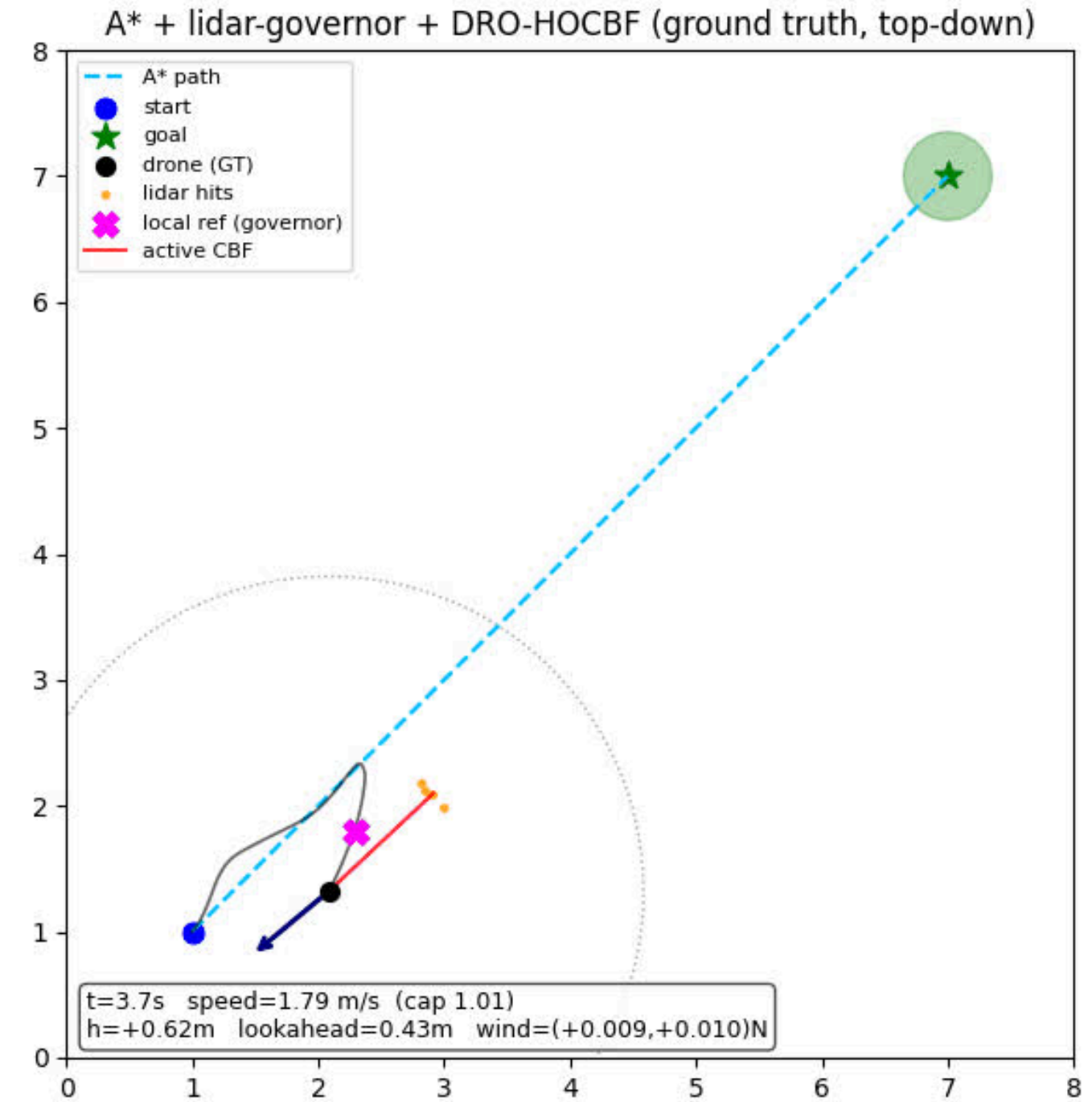
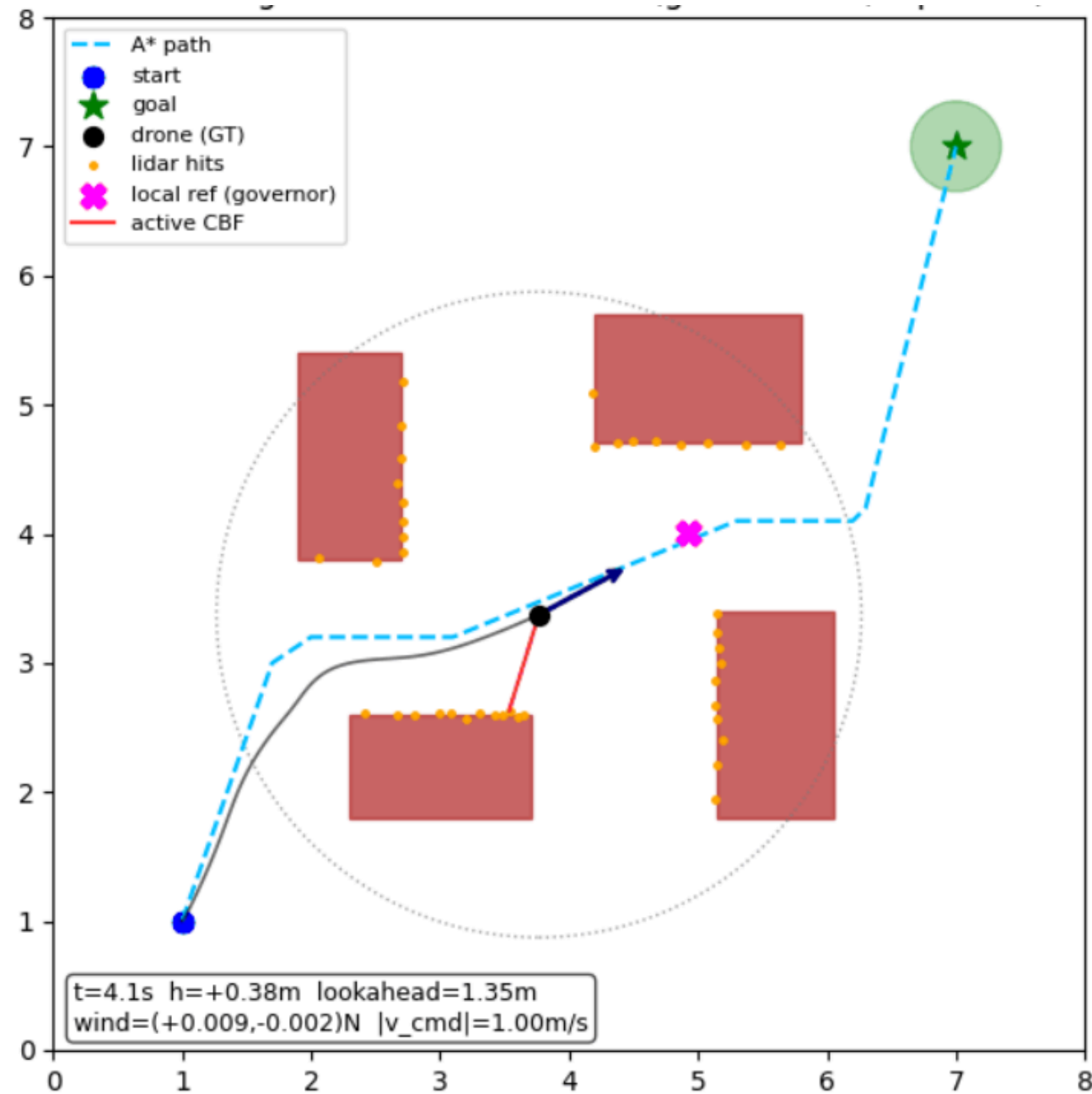
Partial Observation:

The agent (UAV) has partial observation to the environment: $o_t \neq s_t$

We equipy the drone with perception sensors (e.g: LiDAR like livox Mid-360 range or Flow Deck for Crazyflie)



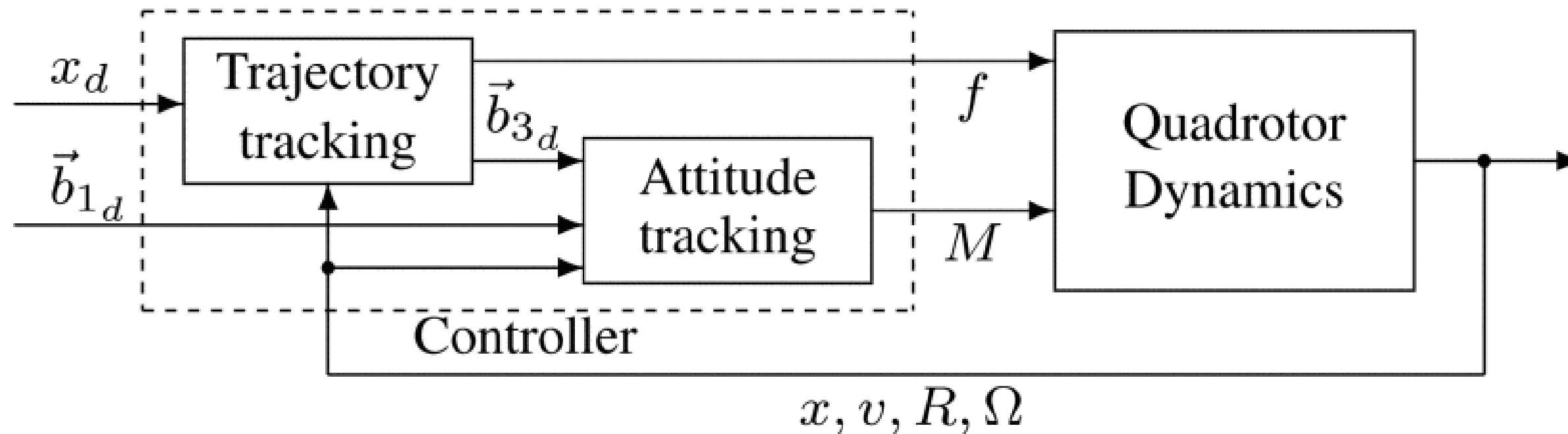
Animation Sample:



Low Level Controller:

We opt to use the geometric low level controller introduced in Lee et al. (2010) which gives very good maneuver to the UAV.

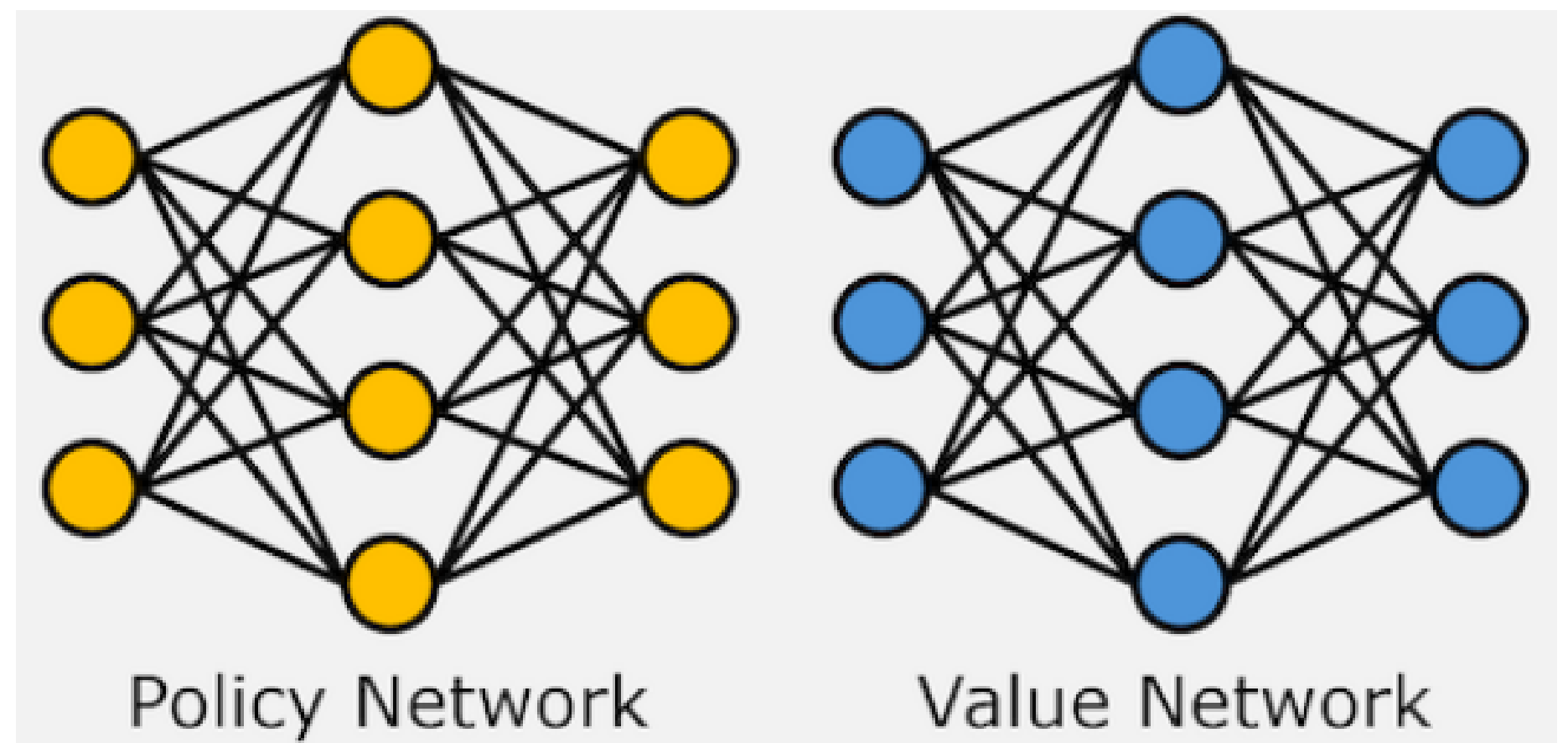
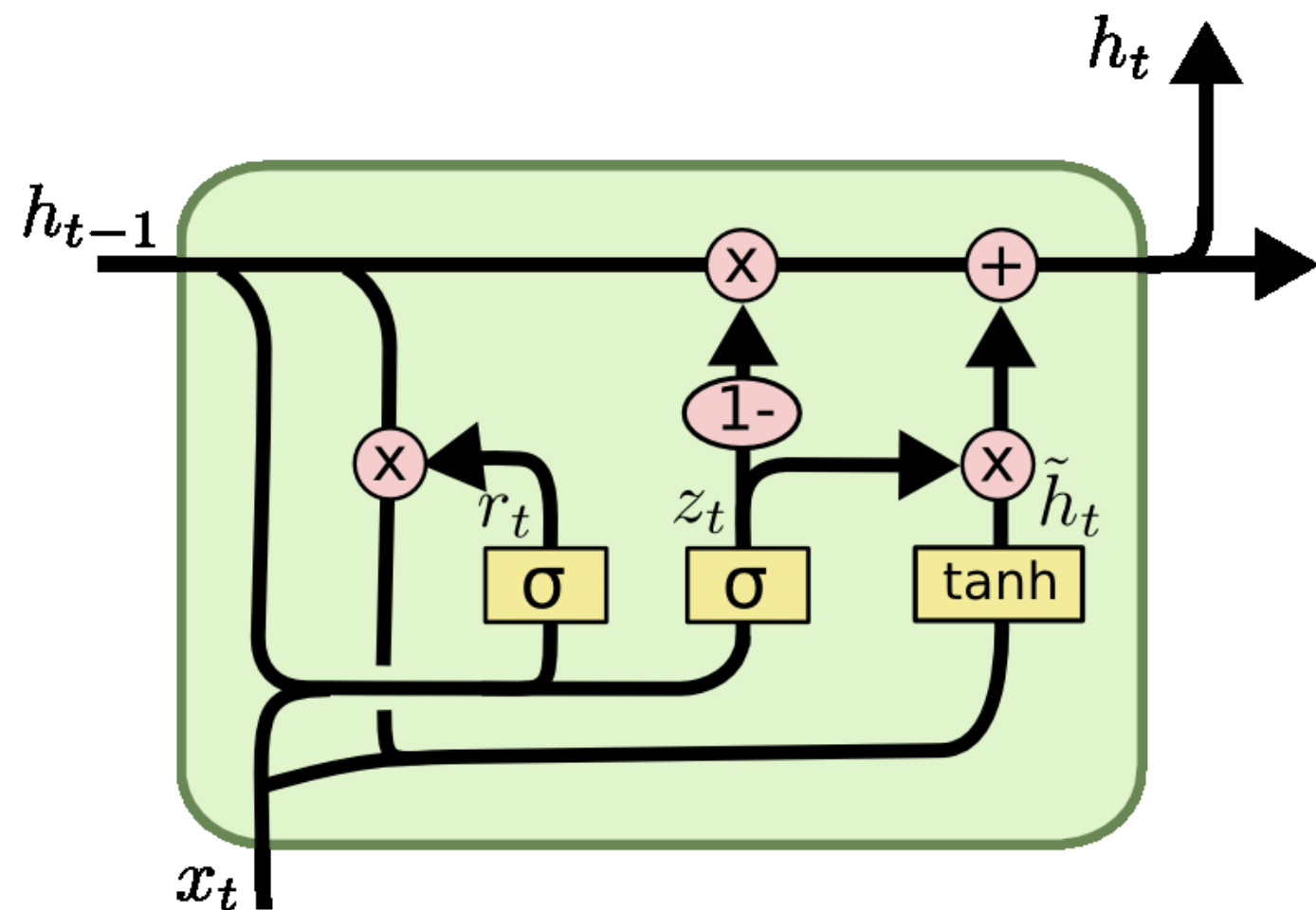
It's implemented in Crazyflie.



Recurrent RL:

Due to the fact that we have only partial observation of the environment, we reformulate our work as Partial Observation Markov Decision Process (PODMP).

A popular approach to mitigate this issue is to use Recurrent RL methods.



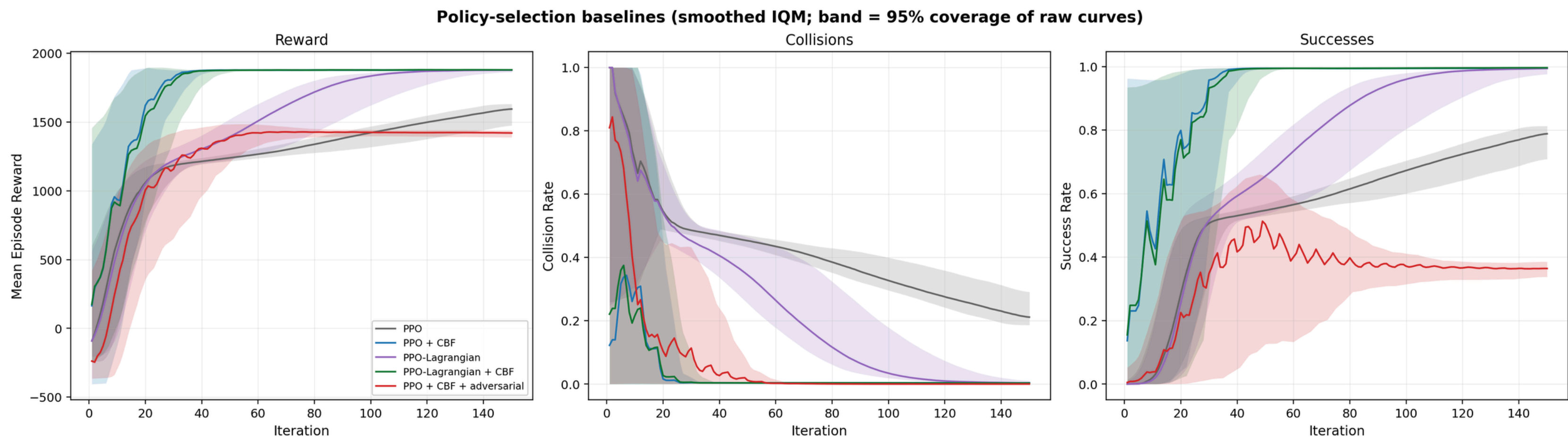
RL methods:

We try different safe RL settings for our work.

Which are vanilla PPO, PPO+CBF, PPO+Lagrangian, PPO + Lagrangian+CBF and PPO + CBF+Adversarial Learner.

The action was sampled from the Beta distribution as it was showed to give better results according to PMLR 70:834-843, 2017.

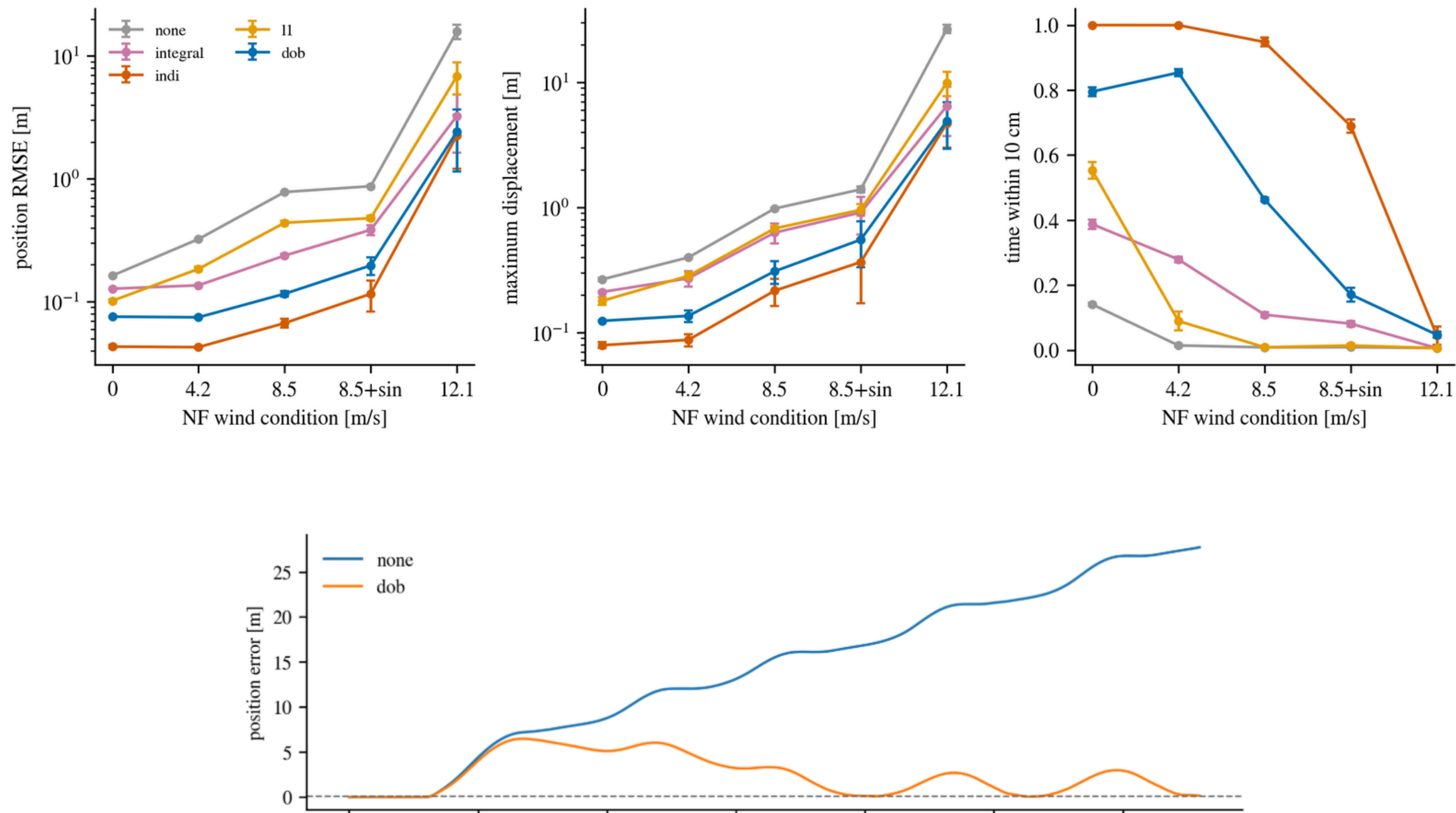
5 seeds were trained for each model

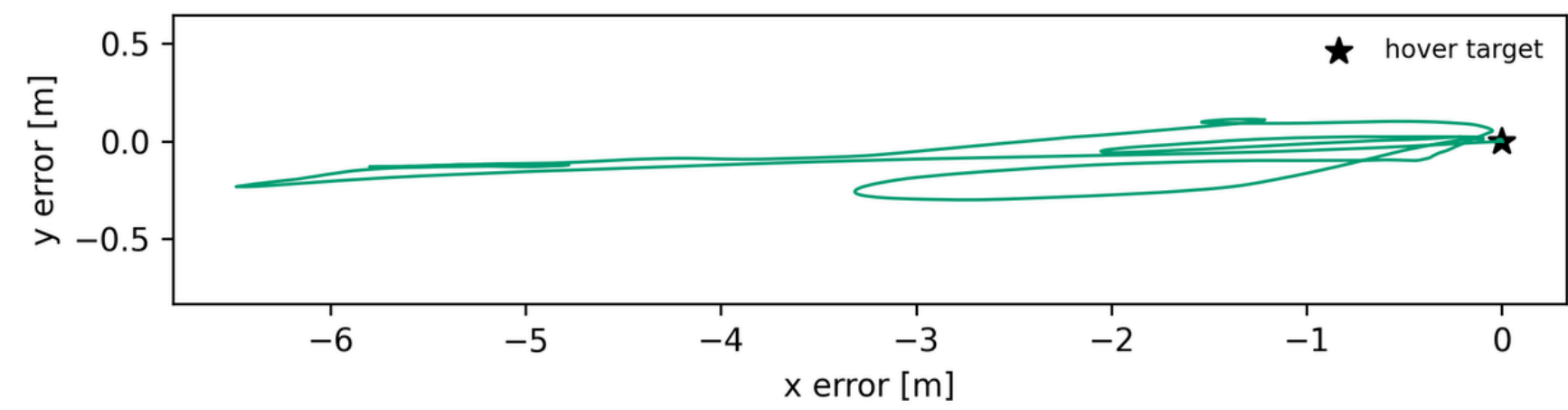
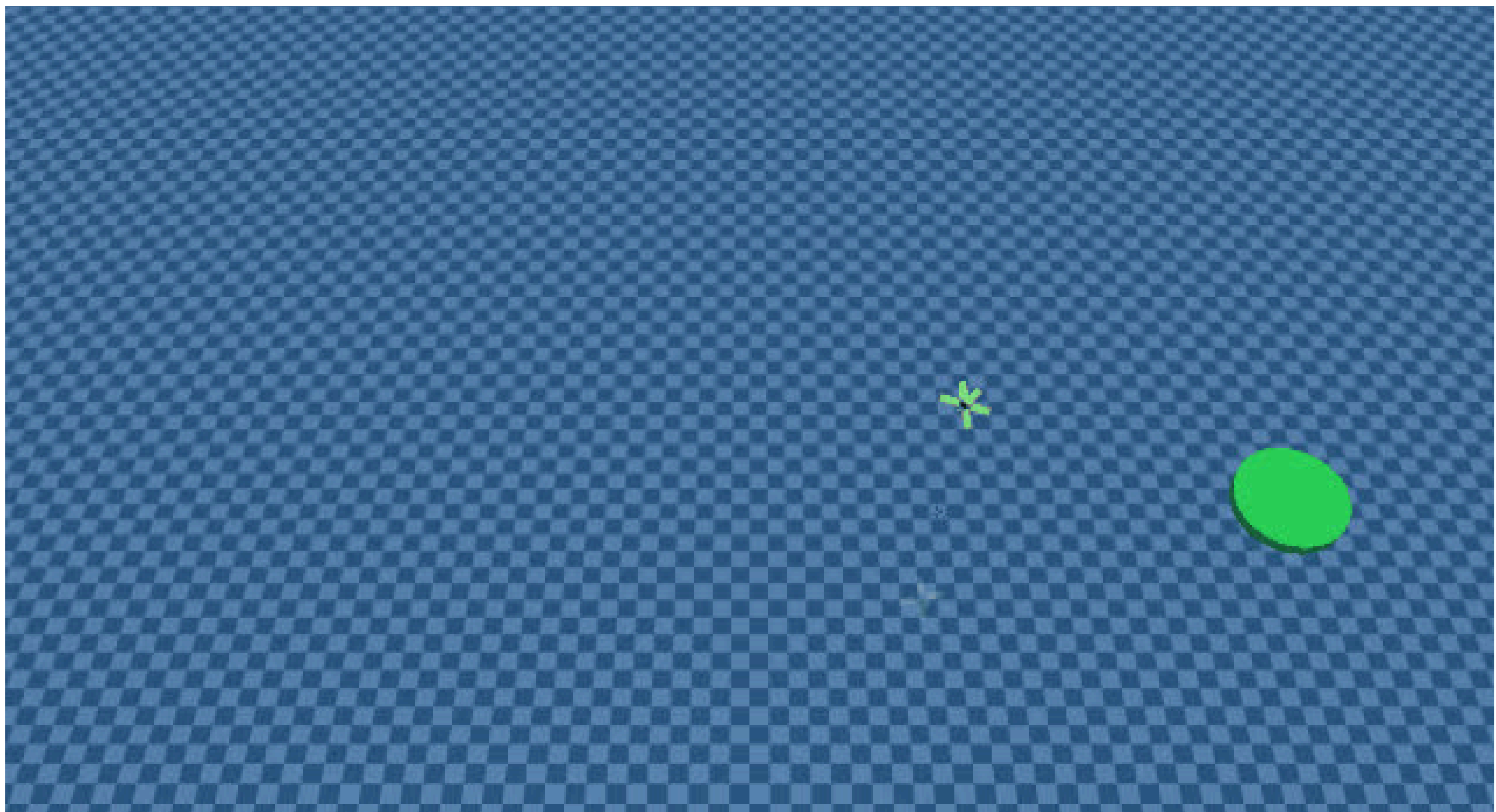


Results:

Environment	Low-complexity		Mid-complexity		High-complexity	
	SR [%]↑	TE↓	SR [%]↑	TE↓	SR [%]↑	TE↓
Static	100.0	0.972	100.0	1.165	100.0	1.349
Dynamic	98.0	1.265	88.0	1.586	67.0	1.952

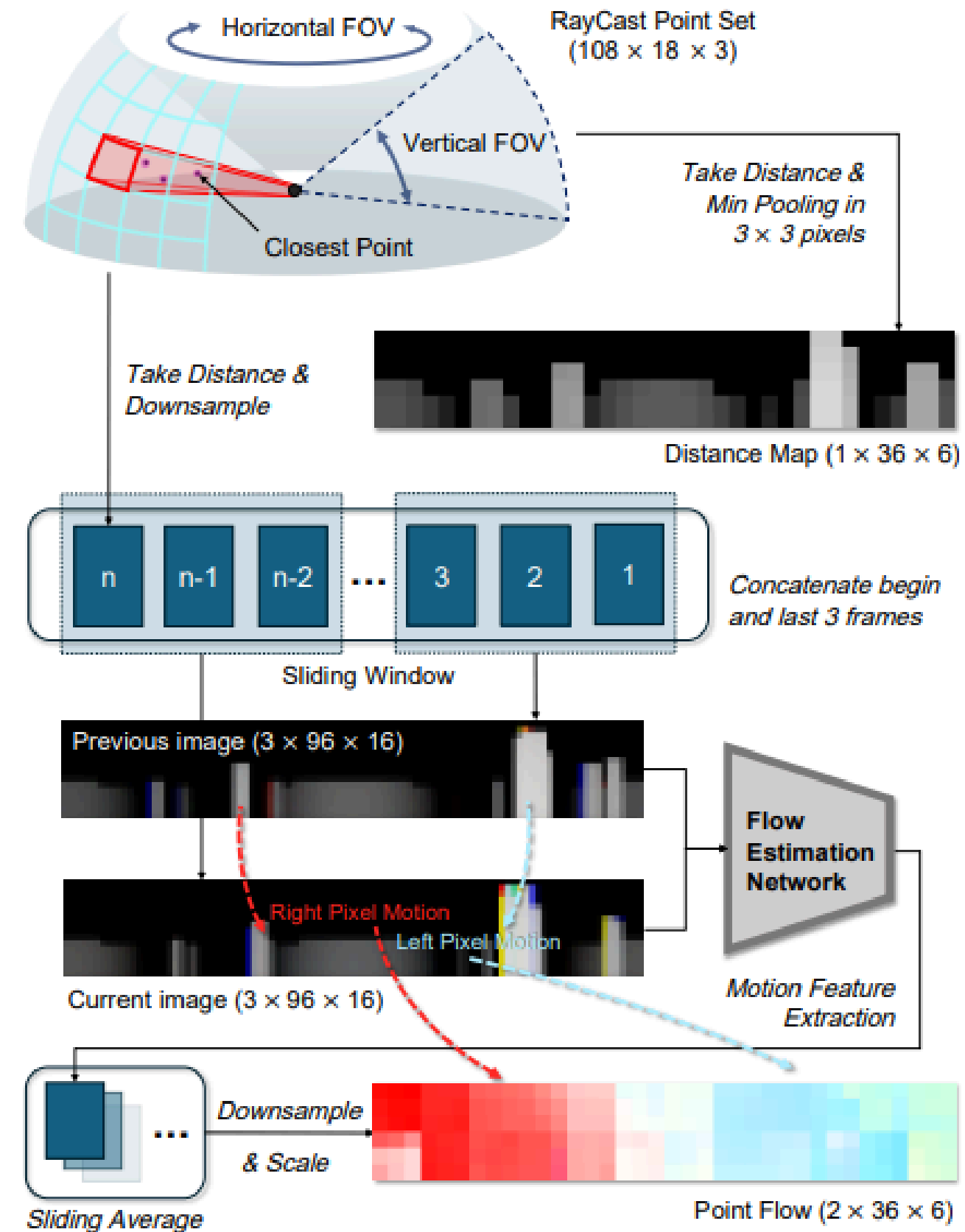
NeuralFly Data: (Science Robotics 2022)



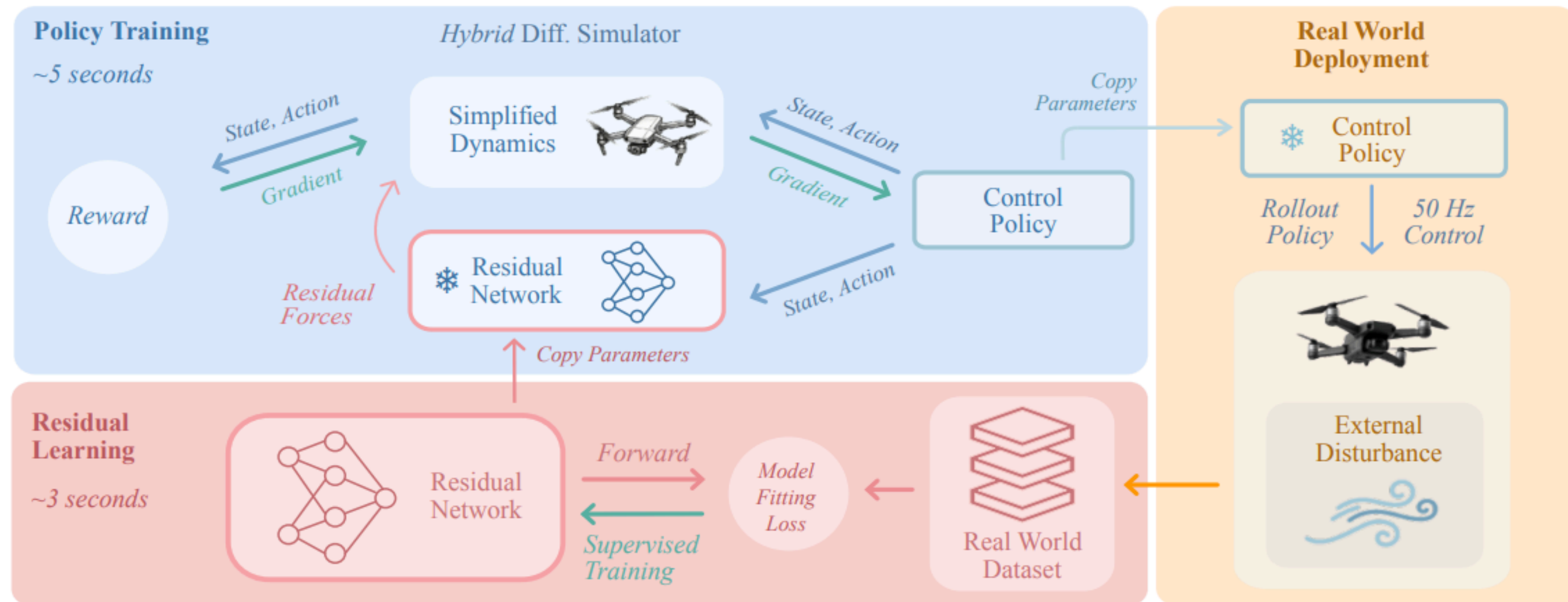


Point-Flow Estimation:

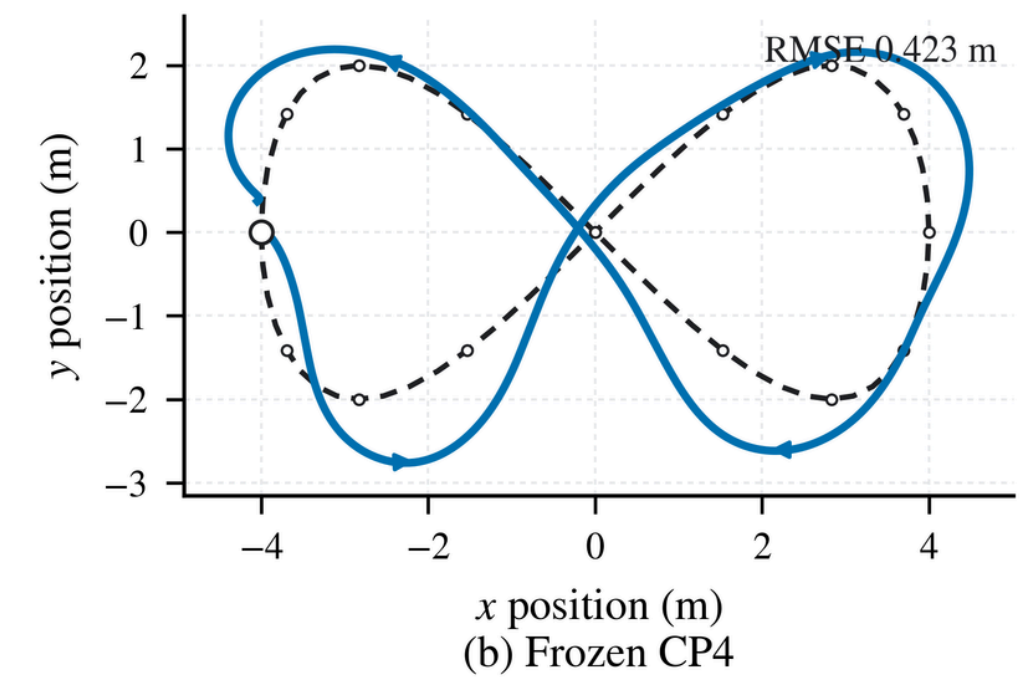
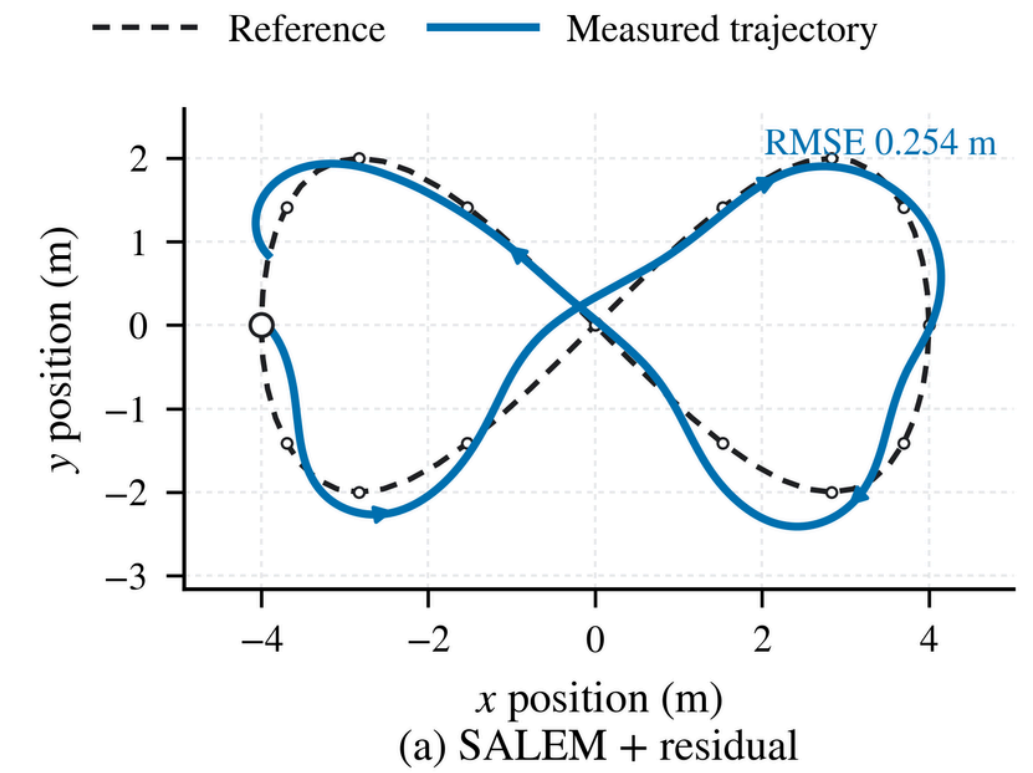
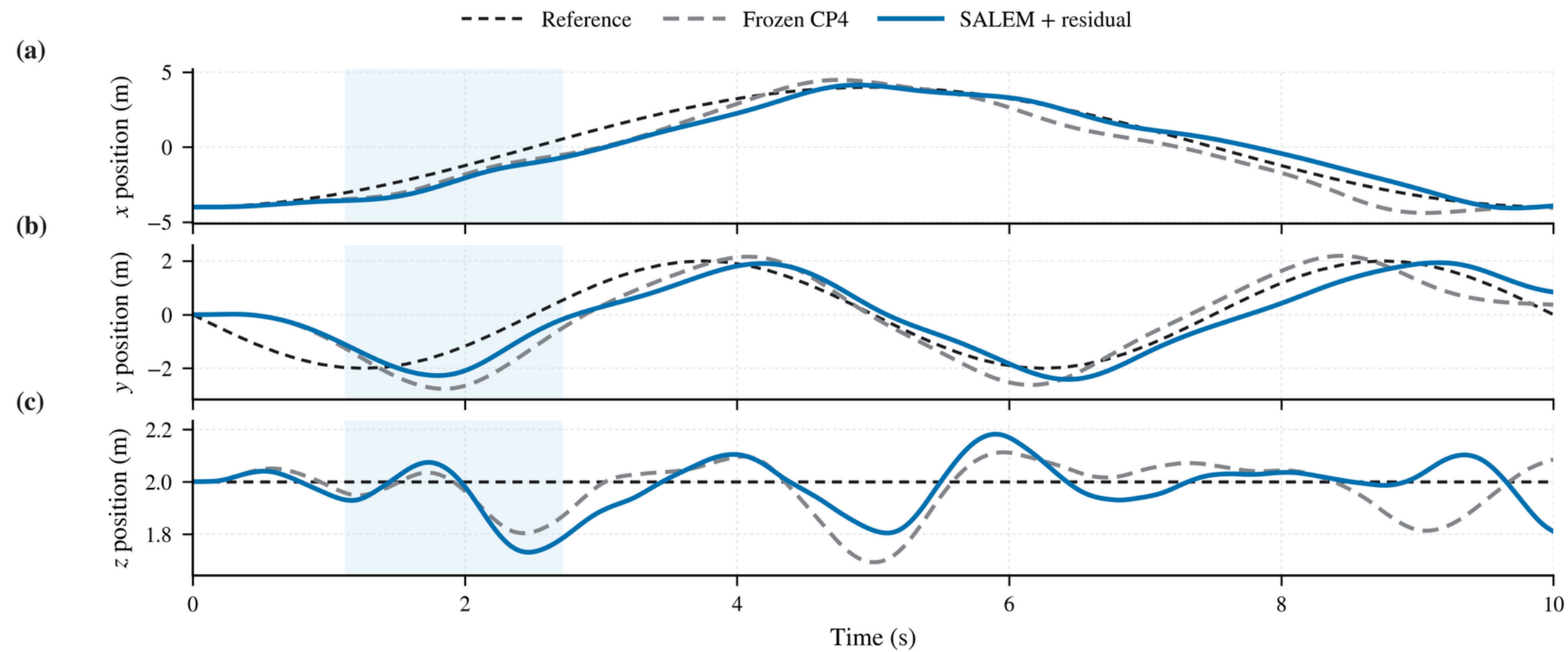
- **Goal:** Extract dynamic motion directly from raw LiDAR data.
- **Input:** Sequential 3D RayCast Point Sets.
- **Process:**
 - Compress to 2D distance maps.
 - Stack past and current frames.
 - Process through a Flow Estimation Network.
- **Output:** A lightweight Point-Flow map detailing environment velocity.
- **Advantage:** Fast, model-free dynamic obstacle awareness.



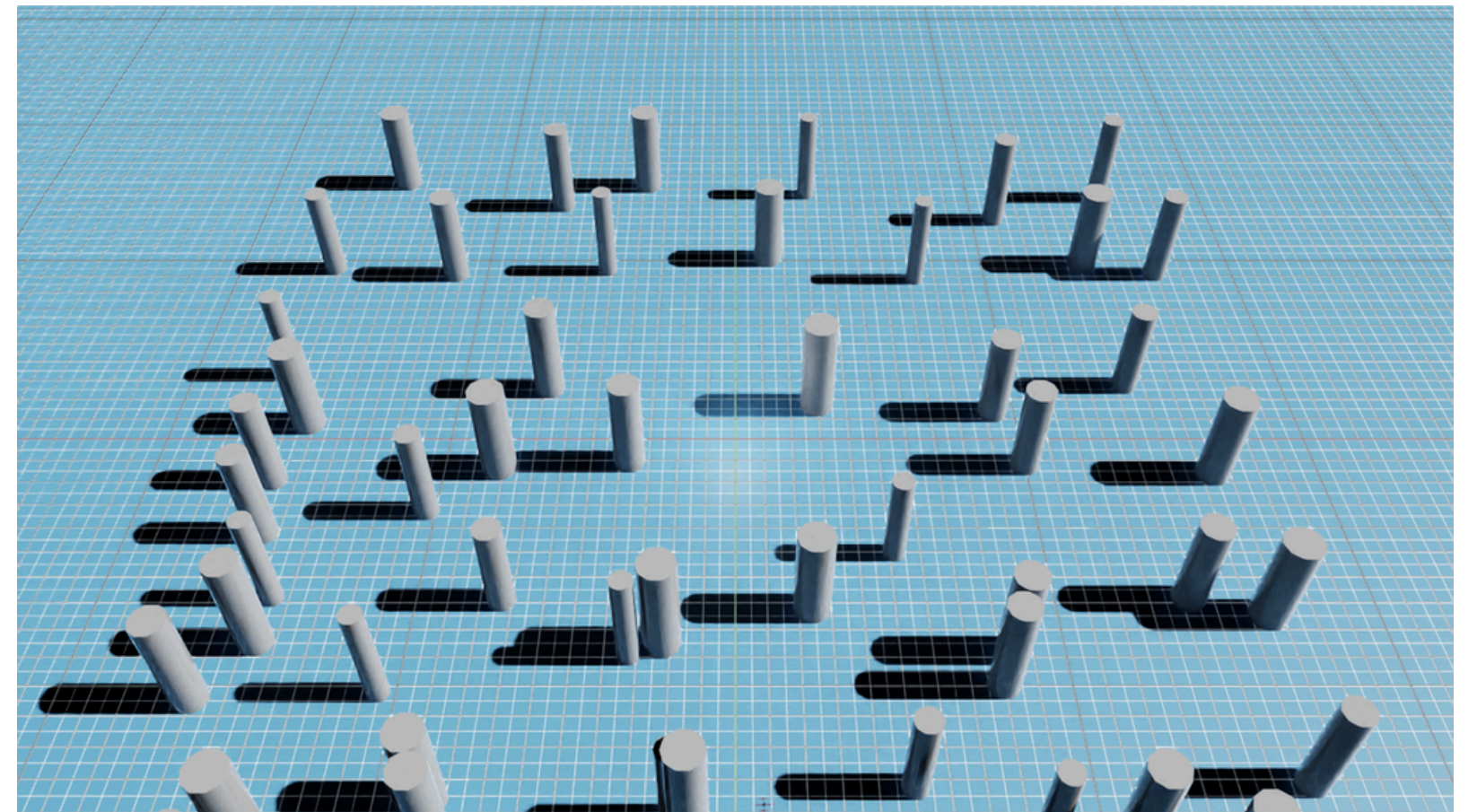
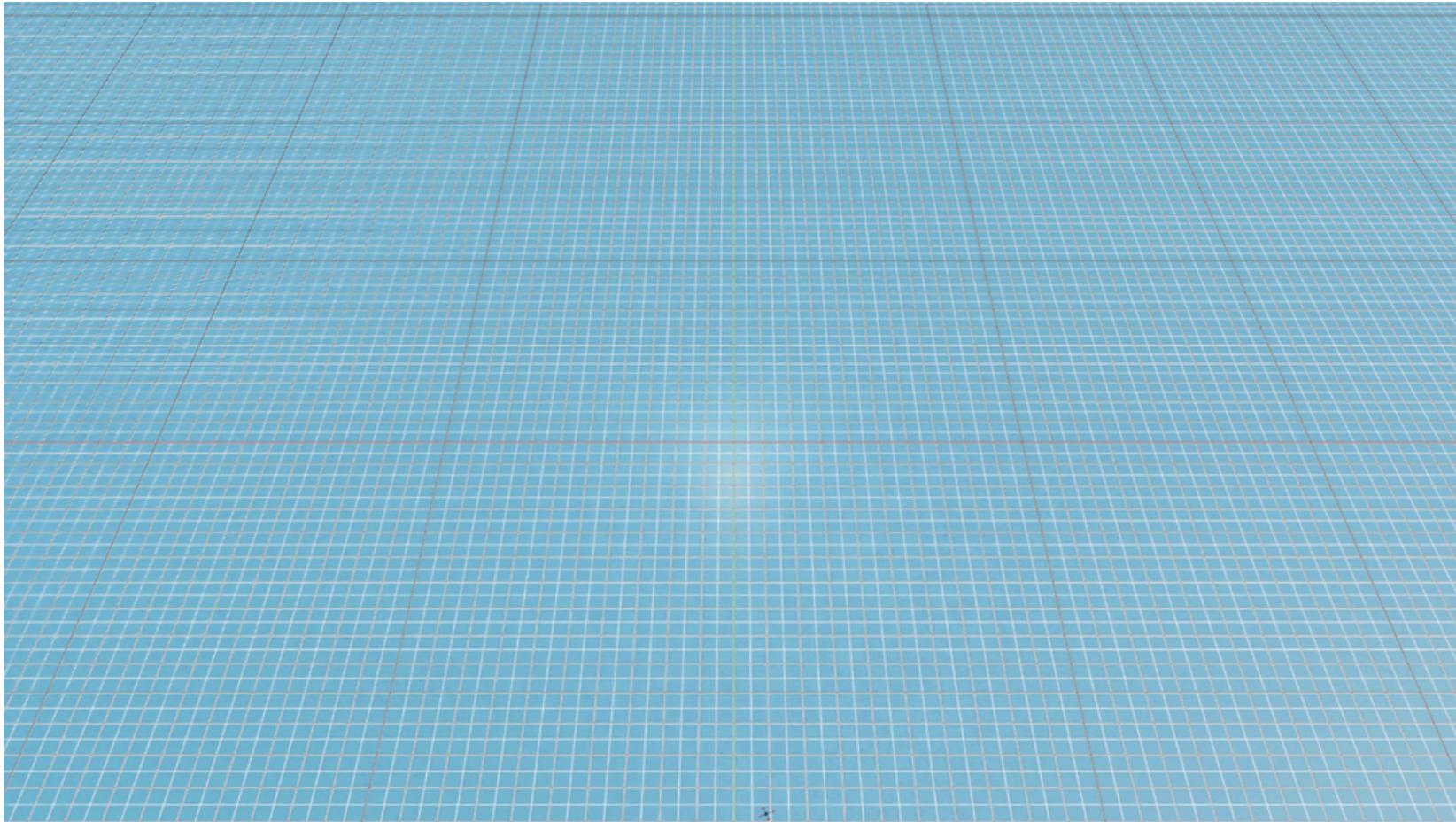
Residual Learning:



Trjactory Tracking:



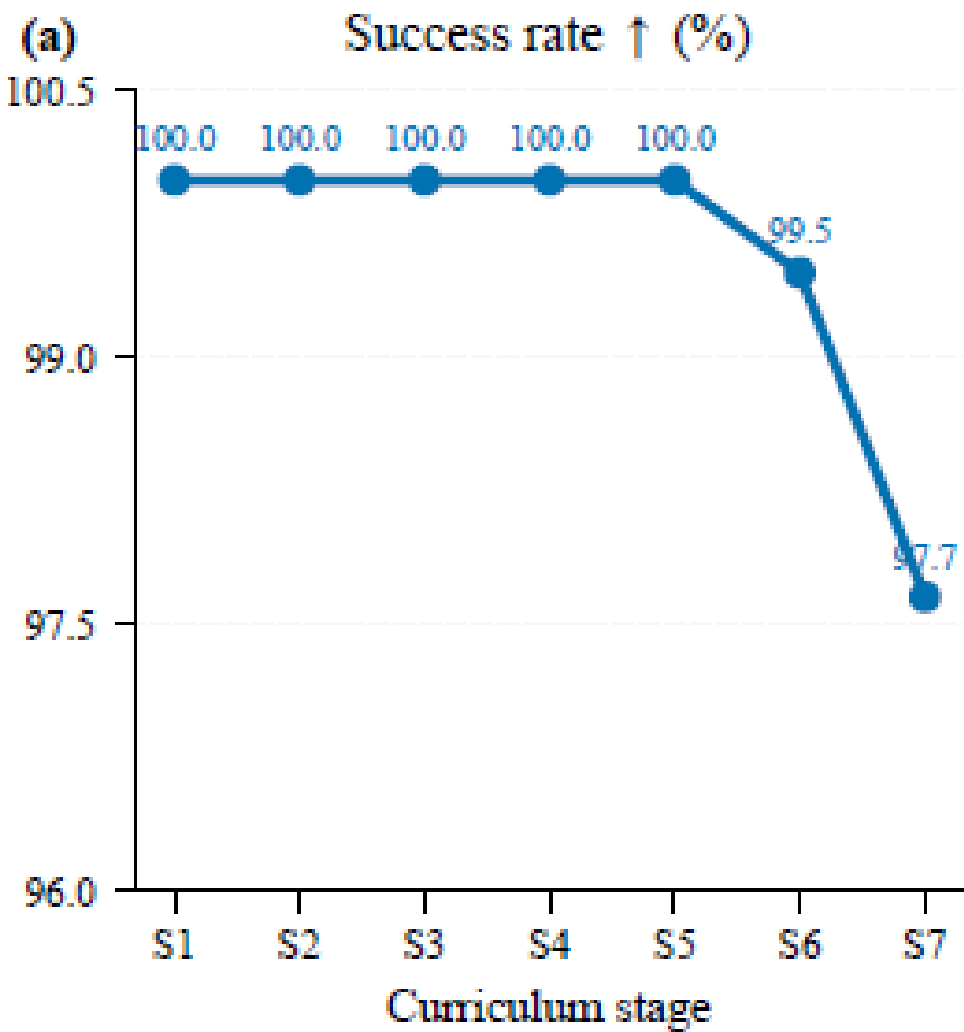
Curriculum Data



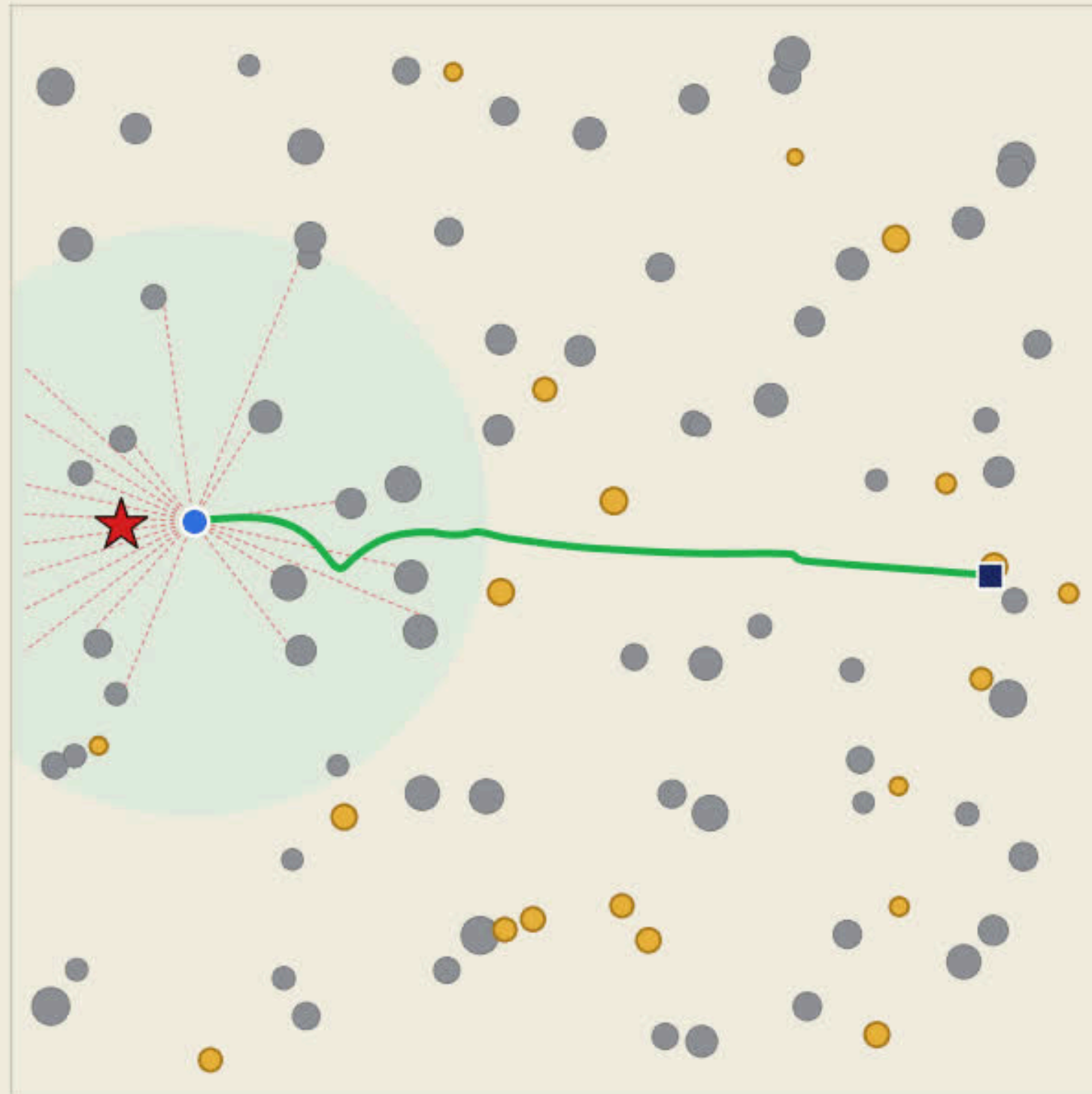
Curriculum Learning & Benchmark

add another experiment fixing the number of obstacles and vary the speed

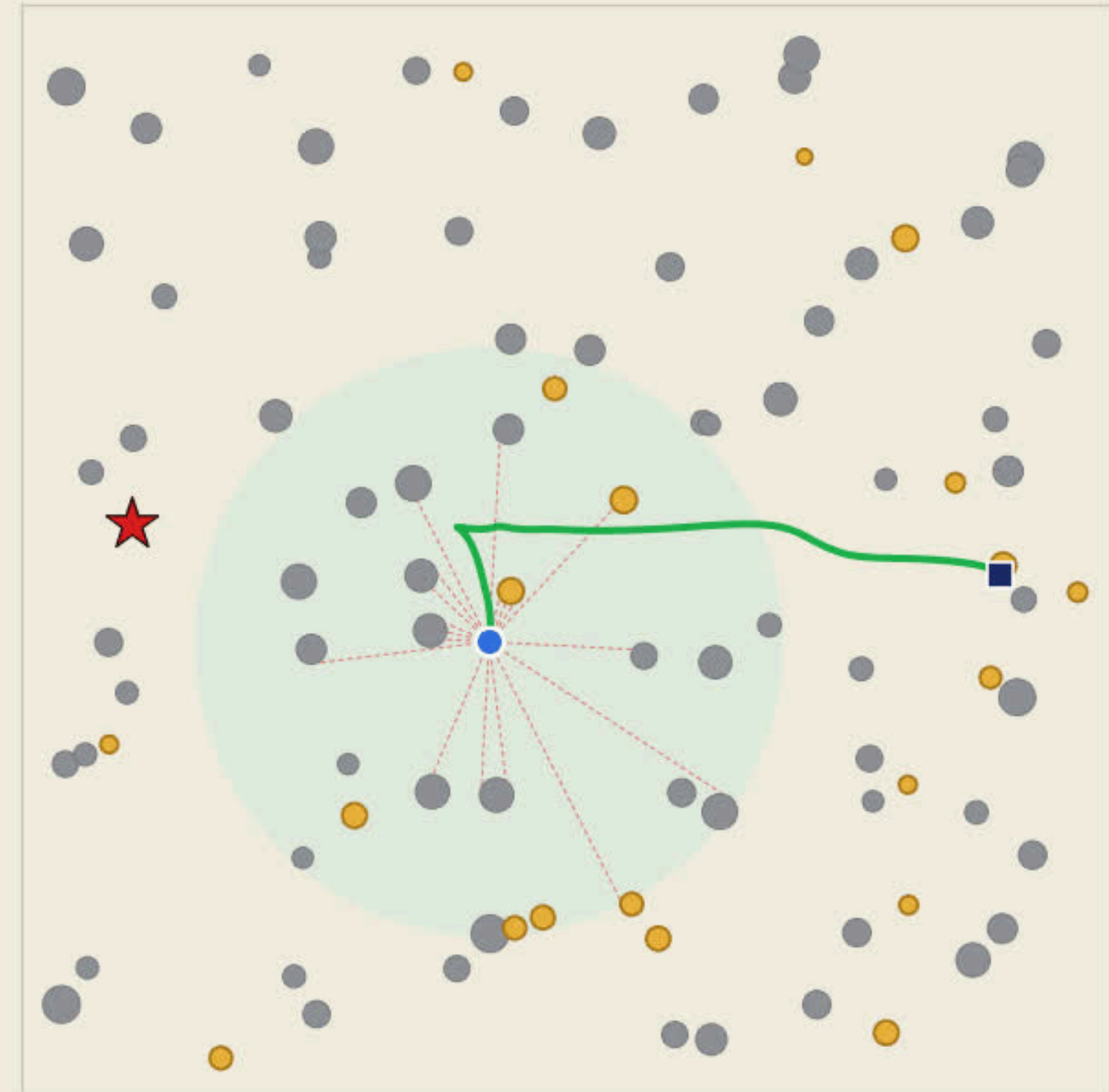
Scenario	Dynamic speed	Method	SR (%)	CR (%)	XTE (m)
72S + 12D	1.0–2.5 m/s	SALEM	99.5	0.5	0.47
		NavRL	74.1	15.4	1.23
72S + 20D	1.0–3.5 m/s	SALEM	93.1	6.6	0.80
		NavRL	61.6	35.1	1.15
72S + 40D	1.0–5.0 m/s	SALEM	32.1	65.6	1.63
		NavRL	23.7	76.3	0.94



SALEM (CP4) — reaches goal



MPC — collision



— flown trajectory ● drone ● dynamic obstacle ★ goal

